École Polytechnique

Detecting PIK3CA mutation in Breast Cancer from Histopathological Slides

OWKIN Machine Learning Challenge

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Abstract

This report presents our approach to the OWKIN Machine Learning Challenge, a weakly supervised binary classification task aiming to predict PIK3CA mutation status from histopathological whole-slide images represented as bags of pre-extracted MoCo features. The dataset comprises 344 training patients and 149 test patients originating from different hospital centers, creating a challenging setting characterized by small sample size, high dimensionality, and pronounced domain shift.

We systematically explored multiple aggregation strategies and modeling approaches, including mean and max pooling, extended statistical summaries, regularized linear models, gradient boosting, attention-based Multiple Instance Learning (MIL), anomaly detection, and ensemble methods. Our best public leaderboard performance (AUC = 0.6554, 55th rank) was achieved with a simple Optuna-tuned Random Forest applied to mean-pooled features.

Our experiments provide empirical evidence that the mutation signal is weak, semi-localized, and variance-dominated under domain shift. The results suggest that the PIK3CA mutation signal is weak and moderately diffuse, rather than being driven by a few highly discriminative tiles. Increasing model complexity or expanding feature dimensionality systematically improved cross-validation scores but degraded generalization performance, revealing variance dominance in this low-n, high-d regime.

We further demonstrate that domain shift across hospital centers constitutes the primary bottleneck for generalization. Cross-validation strategies that fail to simulate this shift yield systematically optimistic estimates, and overly aggressive domain-invariant feature pruning may inadvertently remove genuine biological signal.

Overall, this study highlights that, under severe data scarcity and distribution shift, parsimonious models with implicit variance control can outperform more expressive methods. The findings emphasize the importance of understanding signal structure and validation design rather than relying solely on model complexity.

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1 Introduction

Medical and Scientific Context

Breast cancer is one of the most common cancers worldwide. Modern treatment strategies increasingly rely on molecular characteristics of the tumor, in addition to its visual appearance under the microscope.

One important molecular alteration is the *PIK3CA* mutation. This mutation affects the PI3K signaling pathway, which regulates cell growth and survival. It is present in a significant proportion of breast cancer patients and may influence prognosis and response to targeted therapies.

In clinical practice, the presence of a *PIK3CA* mutation is determined using genetic sequencing techniques. However, these techniques are costly and not always available. Being able to predict the mutation directly from routine histopathology images would therefore be valuable, especially in low-resource settings.

Whole-Slide Images (WSIs) are high-resolution digital scans of tumor tissue. They contain rich morphological information but are extremely large and complex. The challenge is to determine whether subtle visual patterns in these images are associated with the presence of a *PIK3CA* mutation.

Problem Formulation

The task is formulated as a binary classification problem at the patient level. For each patient, the goal is to predict the presence (1) or absence (0) of a *PIK3CA* mutation using histopathology data.

Each patient is represented by approximately 1,000 image tiles extracted from a Whole-Slide Image. However, labels are available only at the patient level. We do not know which specific regions of the tissue may carry the mutation-related signal.

This defines a weakly supervised learning setting. More precisely, the problem follows a Multiple Instance Learning (MIL) structure: each patient corresponds to a bag of tiles, and only the bag is labeled. The model must therefore learn how to aggregate tile-level information into a single patient-level prediction.

Challenge-specific constraints

Several structural characteristics make this task particularly difficult.

Small sample regime. The training set contains only $n = 344$ patients, while the test set contains $n = 149$ patients. Learning reliable patterns from a limited number of samples while avoiding overfitting is a central challenge.

High dimensionality. Each tile is represented by a 2048-dimensional feature vector extracted using a pre-trained neural network. As a result, each patient is described by a 1000×2048 feature matrix before aggregation. Even after summarizing tile-level features, the resulting representation remains high-dimensional relative to the number of patients.

Domain shift. The data originate from five hospital centers. The training set contains slides from centers $\{C1, C2, C5\}$, while the test set contains slides from centers $\{C3, C4\}$. Differences in staining protocols, scanners, and patient populations may introduce systematic variations. The model must therefore generalize to unseen institutions.

MIL structure. Because labels are assigned at the patient level and not at the tile level, only a subset of tiles may contain diagnostically relevant information. Designing an appropriate aggregation strategy is therefore critical.

Evaluation Metric

Model performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

The AUC measures the ability of the model to rank positive patients above negative patients, independently of any classification threshold. Formally,

$$\mathrm{AUC} = \frac{1}{n_+ \cdot n_-} \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{N}} \mathbb{1}\left[\hat{y}_i > \hat{y}_j\right],$$

where $\mathcal{P}$ and $\mathcal{N}$ denote the sets of positive and negative patients, and $\hat{y}_i$ is the predicted probability for patient i.

An AUC of 0.5 corresponds to random guessing, while an AUC of 1.0 indicates perfect ranking. This metric is appropriate in this setting because it is robust to class imbalance and focuses on ranking quality rather than absolute probability calibration.

2 Dataset exploration and preprocessing

Data structure

The dataset consists of:

[nosep]

- **MoCo features:** For each patient, a NumPy array of shape $(1000, 2051)$, where the first 3 columns correspond to tile spatial coordinates $(x, y, \text{zoom level})$ and the remaining 2048 columns are MoCo v2 feature embeddings.
- **Raw tile images:** JPEG images ($\sim$224$\times$224 pixels) organized in patient-specific folders (e.g., `ID_001/ID_001_tile_000_17_170_43.jpg`), with approximately 1000 tiles per patient.
- **Metadata:** `train_metadata.csv` and `test_metadata.csv` containing `Sample ID`, `Center ID`, and `Patient ID` for each slide.
- **Labels:** `train_output.csv` with binary labels (`Target` $\in \{0, 1\}$).

Exploratory analysis

Class distribution. The training set contains 344 samples with 128 positive labels (37.2%), yielding a moderately imbalanced dataset. This imbalance is manageable and does not require aggressive resampling strategies.

Center distribution and domain shift. A critical observation is the **center-level split**:

Table 1: Distribution of samples across hospital centers.

Center ID	Split	Samples
C_1	Train	138
C_2	Train	92
C_5	Train	114
C_3	Test	76
C_4	Test	73

This complete separation of centers between train and test is the primary source of difficulty. Any model that inadvertently learns center-specific "signatures" (e.g., staining intensity, scanner artifacts) will fail to generalize. We confirmed this risk through adversarial validation experiments (Section 6.2).

Feature statistics. The MoCo features exhibit high redundancy: PCA analysis reveals that **90% of variance** is captured by 97 components, while 95% requires 149 components (out of the 344 maximum given $n < d$). Some feature dimensions show near-zero variance (constant across tiles and patients), while others exhibit high skewness and kurtosis, particularly when computing higher-order statistics.

Preprocessing pipeline

Our preprocessing involves:

[nosep]

1. **Coordinate removal:** Discarding the first 3 columns (spatial coordinates) from the MoCo features, retaining only the 2048-dimensional embeddings.

2. **NaN handling:** Computing skewness and kurtosis involves dividing by the standard deviation σ. When a feature dimension has zero variance across all 1000 tiles ($\sigma = 0$), these statistics are undefined, producing NaN or $\pm\infty$. Replacing them with **zero** is mathematically justified: a constant distribution has zero skewness (perfect symmetry) and zero excess kurtosis (no tails). We verified that no NaN values originated from other sources.

3. **Standardization:** Applying `StandardScaler` fitted on the training set only (for PCA-based pipelines) to prevent data leakage.

3 Feature engineering

Given the MIL structure of the data (1000 tiles $\times$ 2048 features per patient), a fundamental design choice is **how to aggregate tile-level features into a patient-level representation**. We explored multiple strategies.

Mean pooling (baseline)

The simplest aggregation: compute the element-wise mean across all 1000 tiles, producing a 2048-dimensional patient-level feature vector.

$$\mathbf{x}_{\text{patient}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{z}_i \in \mathbb{R}^{2048} \tag{1}$$

where $\mathbf{z}_i \in \mathbb{R}^{2048}$ is the MoCo embedding of tile i. This naive approach discards all distributional information but proved remarkably effective, likely because the MoCo encoder already produces well-structured representations.

Extended statistics (mean, std, max)

To capture distributional properties of the tile population, we extended the aggregation to include:

$$\mathbf{x}_{\text{ext}} = [\mu(\mathbf{z}),\ \sigma(\mathbf{z}),\ \max(\mathbf{z})] \in \mathbb{R}^{6144} \tag{2}$$

The standard deviation captures intra-slide heterogeneity (a hallmark of aggressive tumors), while the maximum captures "hotspot" tiles that may be most diagnostically relevant.

Full statistical profile (9 statistics)

We further expanded the feature set to 9 statistics per dimension:

$$\mathbf{x}_{\text{full}} = [\mu,\ \sigma,\ \max,\ \min,\ Q_{25},\ Q_{50},\ Q_{75},\ \text{skew},\ \text{kurt}] \in \mathbb{R}^{18432} \tag{3}$$

This produces 18,432 features per patient, providing a comprehensive distributional fingerprint. However, this also dramatically increases the risk of overfitting given $n = 344$.

4 Understanding the Nature of the Mutation Signal

A central question of this project is not only how to predict the mutation, but what kind of signal the mutation induces in histopathological images. Understanding whether the *PIK3CA* mutation manifests as a strong localized pattern or as a subtle global alteration is essential to design appropriate aggregation and modeling strategies.

Localized versus Diffuse Signal Hypothesis

We formulated two competing hypotheses regarding the spatial structure of the mutation signal.

Localized hypothesis. If the mutation generates highly discriminative morphological patterns confined to a small subset of tiles ("hotspots"), then max pooling or top-k pooling should be effective. Under this assumption, only a few tiles would carry strong predictive information, and capturing the highest activation values would be sufficient for patient-level prediction.

Diffuse hypothesis. Alternatively, the mutation may induce subtle but widespread changes in tissue morphology. In this case, no single tile would be strongly discriminative, but the overall distribution of tile features would shift slightly between mutated and non-mutated patients. Mean pooling would then be more appropriate, as it captures global distributional differences.

Empirically, mean pooling consistently outperformed strict max pooling strategies, while extended statistics (e.g., skewness, kurtosis, high-order moments) did not yield stable improvements on the leaderboard. This suggests that the signal is not purely sparse and extreme, but neither is it uniformly distributed across all tiles.

Evidence from Model Behavior

The behavior of different models further informs our understanding of the signal structure.

First, increasing model complexity (gradient boosting, deep attention-based MIL) did not significantly improve performance over a moderately regularized Random Forest applied to mean-pooled features. If the mutation signal relied on highly non-linear interactions between tiles, more expressive models would be expected to provide substantial gains.

Second, expanding the feature space from 2048 dimensions (mean pooling) to 6144 or 18,432 dimensions (extended statistics) improved cross-validation scores but degraded leaderboard performance. This indicates that richer distributional summaries may capture center-specific artifacts rather than robust biological patterns.

Taken together, these observations suggest that the mutation signal is weak, moderately structured, and primarily encoded in global shifts of the tile feature distribution rather than in rare, extreme activations.

Interpretation of Tile-Level Probability Distributions

To further investigate this hypothesis, we analyzed tile-level probability distributions (see Section 6.3). Tiles from mutated patients exhibit a heavier right tail compared to non-mutated patients, indicating the presence of moderately high-probability regions (Figure 1).

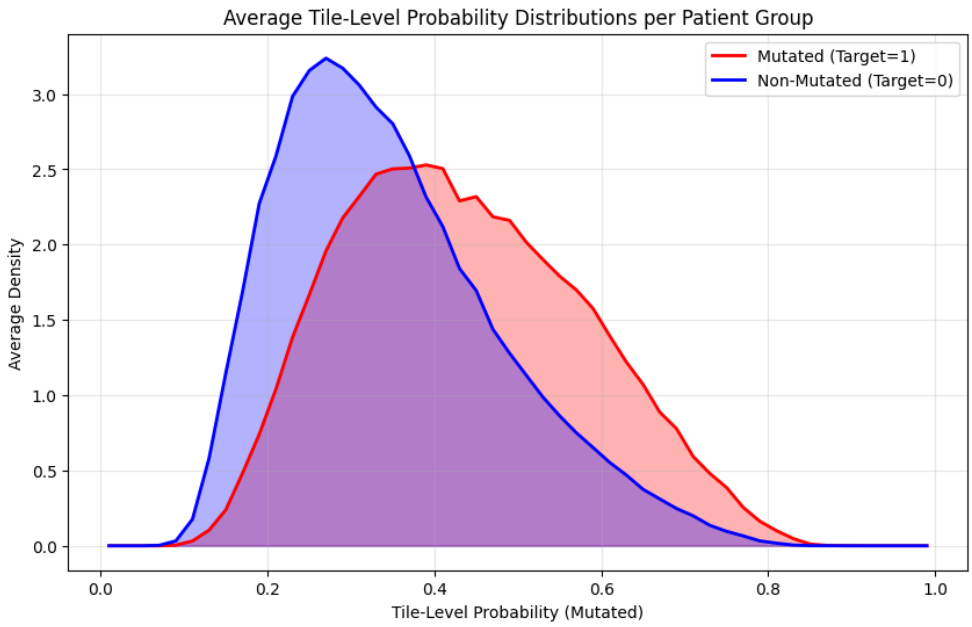

Figure 1: Average Tile-Level Probability Distributions per Patient Group. The mutated group shows a heavier right tail, indicating slightly elevated probabilities across subsets of tiles rather than a few highly discriminative regions.

However, these high-probability tiles are not sufficiently dominant to allow pure max pooling to outperform mean aggregation. Instead, their contribution appears to shift the average tile-level prediction upward, explaining why mean pooling combined with a variance-reducing model such as Random Forest performs robustly.

Overall, the mutation signal appears to be semi-localized but weak: it manifests through a combination of slightly elevated probabilities across subsets of tiles rather than through singular, highly discriminative regions.

This characterization provides a unifying explanation for our empirical findings and guides the interpretation of subsequent modeling results.

5 Cross-validation strategy

Designing a reliable cross-validation (CV) procedure was one of the main methodological challenges of this competition. Because the training and test sets originate from different hospital centers, standard validation strategies may produce overly optimistic performance estimates.

Why Standard Cross-Validation Overestimates Performance

We first employed StratifiedKFold (k = 5), which preserves class balance across folds. However, this strategy mixes samples from the same hospital centers across training and validation splits. The model is therefore evaluated on data drawn from the same domain distribution it was trained on.

In the presence of center-specific artifacts (e.g., staining intensity, scanner characteristics), the model may learn domain signatures rather than purely biological features. This domain leakage leads to systematically optimistic AUC estimates and explains the consistent gap observed between cross-validation scores and public leaderboard performance.

GroupKFold and High Variance under Limited Domains

To better simulate the test scenario, we implemented GroupKFold using hospital center ID as the grouping variable. This ensures that entire centers are held out during validation, approximating an out-of-domain evaluation.

However, the training data contain only three centers ({C1, C2, C5}). GroupKFold therefore operates over only three groups, producing at most three effective validation splits. With such few domains, performance estimates exhibit high variance and depend strongly on which center is held out.

Limitations of Leave-One-Center-Out (LOCO)

A Leave-One-Center-Out (LOCO) strategy would most faithfully reproduce the test setting. Yet, with only three training centers, LOCO also yields three folds, resulting in unstable AUC estimates and unreliable model selection due to small validation sample sizes.

Practical Compromise

Given these constraints, we adopted patient-level StratifiedKFold with multiple repeats as our primary model selection strategy. Although this approach does not fully capture domain shift, it provides lower-variance estimates and more stable hyperparameter optimization.

Overall, this competition highlights a fundamental bias–variance tradeoff in validation under domain shift: domain-aware strategies reduce bias but increase variance when the number of domains is small, whereas standard CV provides stability at the cost of optimistic estimates.

6 Predictive pipeline and model exploration

We followed an iterative experimental approach, progressively exploring more complex methods while maintaining rigorous cross-validation. Figure 2 illustrates our overall pipeline architecture. The project can be interpreted as a structured exploration across three axes: (i) aggregation hypothesis, (ii) modeling hypothesis, (iii) domain generalization strategy.

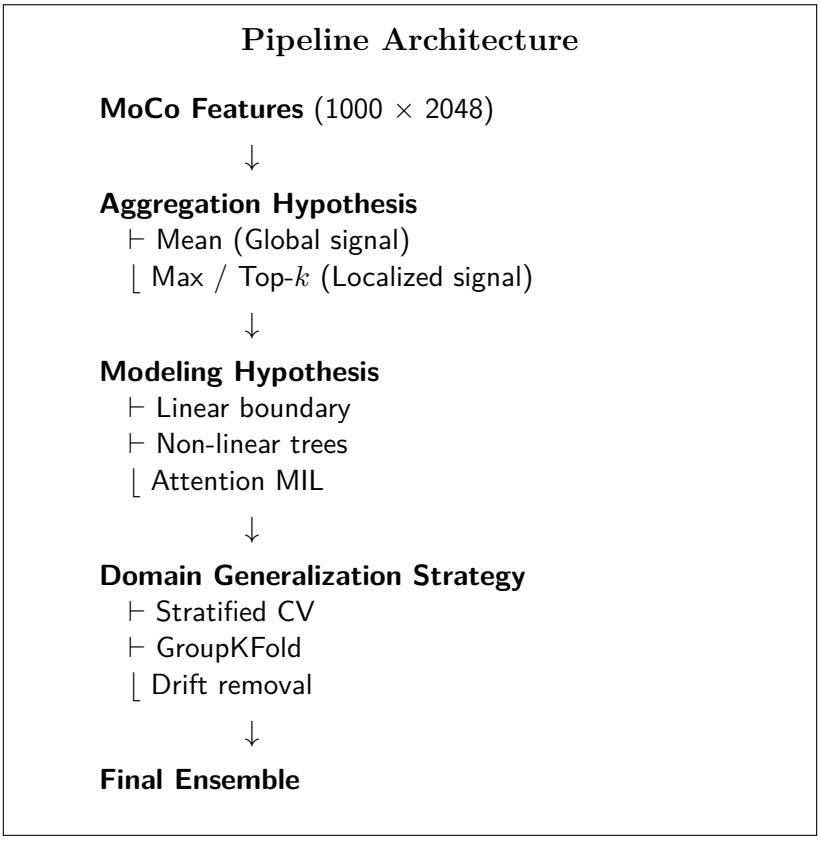

Figure 2: High-level pipeline architecture. We systematically explored different hypotheses across aggregation, modeling, and domain generalization strategies before combining them into a final ensemble.

Iteration 1: Baseline Random Forest + Optuna

Our first approach used **mean-pooled MoCo features** (2048 dimensions) with a **Random Forest classifier** optimized via Optuna [Akiba et al., 2019]. Cross-validation was performed at the **patient level** using `StratifiedKFold` ($k = 5$), ensuring that all slides from the same patient appear exclusively in one fold.

Optuna explored 20 trials over the following search space:

Table 2: Optuna search space for Random Forest baseline.

Parameter	Range
`n_estimators`	$[10, 200]$
`max_depth`	$[2, 30]$
`min_samples_split`	$[2, 20]$
`min_samples_leaf`	$[1, 10]$

Result: Public leaderboard AUC = **0.6554**. This remained our best submission throughout

the challenge.

Key insight: Despite its simplicity, this model benefited from Random Forest's natural resistance to overfitting (bagging + feature subsampling), which proved critical in this low-n, high-d regime.

Iteration 2: XGBoost with feature engineering

We hypothesized that XGBoost's gradient boosting mechanism would capture more complex feature interactions than Random Forest. We enriched the feature set with Mean, Std, and Max pooling (6144 dimensions) and tuned XGBoost via Optuna with 30 trials.

Adversarial validation. To mitigate domain shift, we trained a classifier to distinguish training from test samples. Features with the highest discriminative power (measured by Random Forest feature importance) were flagged as "drifting" and removed. We experimented with two intensities:

[nosep]

- **Light pruning:** Remove top 20 drifting features
- **Aggressive pruning:** Remove top 2000 drifting features (Section 6.6)

Results:

Table 3: XGBoost experiments.

Configuration	CV AUC	Public LB AUC
XGBoost (Mean only)	0.680	0.6185
XGBoost + Mean/Std/Max	0.680	0.6137
XGBoost + Mean/Std/Max + Drift Removal	0.680	0.6346

Observation: XGBoost consistently *overfitted*, with a ∼0.05 gap between CV and leaderboard scores. The additional features (Std, Max) did not improve generalization; they likely introduced center-specific noise that XGBoost exploited. Adversarial validation helped slightly but was insufficient to close the gap.

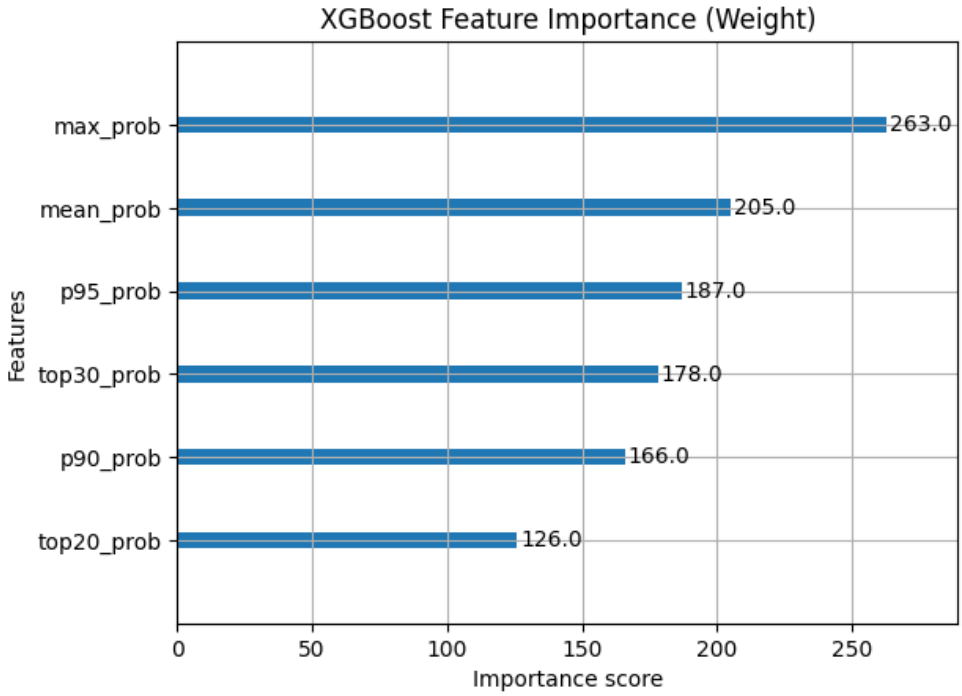

Figure 3: XGBoost Feature Importance (Weight), showing the relative contribution of the top features to the model.

Iteration 3: PCA + regularized linear models

To combat the curse of dimensionality, we applied PCA dimensionality reduction followed by regularized logistic regression. The pipeline:

[nosep]

1. Features: Mean + Std + Max pooling → 6144 dimensions
2. StandardScaler (fit on train only)
3. PCA: grid search over $n \in \{50, 100, 150, 200, 250\}$
4. Model comparison: LogReg L2, LogReg L1, Ridge, ElasticNet
5. Regularization: 9 values of $C \in [10^{-3}, 10]$
6. Ensemble: 25 models (5×5 folds/repeats), averaged

Best configuration: PCA(150) + Logistic Regression L2 with $C = 0.01$, achieving Public LB AUC = **0.6434**.

Analysis: The result was below the RF baseline. PCA captures directions of maximum variance, which are not necessarily the most discriminative. The linear assumption of logistic regression further limits expressiveness. However, PCA+Ridge produced *partially uncorrelated* errors with RF, motivating our ensemble strategy.

Iteration 4: RF + PCA-Ridge ensemble

We blended the Random Forest and PCA-Ridge predictions:

$$\hat{y}_{\text{blend}} = \alpha \cdot \hat{y}_{\text{RF}} + (1 - \alpha) \cdot \hat{y}_{\text{Ridge}} \tag{4}$$

Grid search over α on out-of-fold (OOF) predictions yielded $\alpha^* = 0.55$, with an OOF AUC of **0.6975**, our highest CV score. However, the public leaderboard returned only **0.6306**, a significant degradation indicating that the ensemble overfitted to the OOF distribution.

Table 4: Ensemble blending analysis on OOF predictions.

Method	**OOF AUC**
RF alone (Optuna-tuned)	0.6774
PCA-Ridge alone	0.6932
Simple average (0.5/0.5)	0.6972
Best blend ($\alpha = 0.55$)	0.6975
Stacking (LogReg)	0.6934

Iteration 5: Extended features with conservative RF

We enriched the feature set to 9 statistics (18,432 dimensions) and applied a *constrained* Optuna search space specifically designed to prevent overfitting:

[nosep]

- `max_depth` $\in [3, 8]$ (shallow trees)
- `min_samples_split` $\in [5, 30]$ (high regularization)
- `min_samples_leaf` $\in [5, 25]$ (high regularization)
- `max_features` $\in [0.05, 0.3]$ (aggressive subsampling)

We also applied adversarial validation to remove the top 50 drifting features. CV scores were promising (AUC ≈ 0.69), but we did not observe improvements over the baseline on the leaderboard.

Iteration 6: Domain-agnostic pipeline

Acknowledging that the train/test split is across hospital centers, we implemented a fully domain-agnostic pipeline:

[nosep]

1. **Aggressive adversarial validation:** Remove the top 2000 features most correlated with the domain (train vs. test).
2. **GroupKFold cross-validation:** Use `Center ID` as the group variable, ensuring that entire hospitals are held out during validation. This simulates the real test scenario where the model encounters unseen institutions.

Result: GroupKFold CV AUC = **0.6862**, but Public LB AUC = **0.6241**. This was one of our worst-performing submissions at the time. Removing 2000 features likely eliminated genuinely discriminative biological features along with the center-specific artifacts. Moreover, the extended statistics (skewness, kurtosis) are inherently more sensitive to domain-specific variations in staining and scanning quality, amplifying the problem.

This experiment provided a crucial lesson: *domain invariance pursued too aggressively can destroy the signal itself.*

Iteration 7: Attention-based MIL (deep learning)

We implemented an AB-MIL model [Ilse et al., 2018] operating directly on the pre-extracted MoCo features. Unlike the aggregation-based approaches, AB-MIL learns an **attention mechanism** that assigns importance weights to each tile, then aggregates them via a weighted sum:

$$a_k = \frac{\exp\left\{w^\top \tanh(V\mathbf{z}_k) \odot \sigma(U\mathbf{z}_k)\right\}}{\sum_{j=1}^{N} \exp\left\{w^\top \tanh(V\mathbf{z}_j) \odot \sigma(U\mathbf{z}_j)\right\}} \tag{5}$$

$$\mathbf{z}_{\text{patient}} = \sum_{k=1}^{N} a_k \mathbf{z}_k \tag{6}$$

This "gated attention" mechanism allows the model to focus on diagnostically relevant tiles while ignoring uninformative ones. The aggregated representation $\mathbf{z}_{\text{patient}}$ is passed through a small classifier head.

Architecture: MoCo input (2051d) → Gated Attention (256d hidden) → FC(64) + ReLU + Dropout(0.3) → Sigmoid output.

Training: 5-fold StratifiedKFold, AdamW (lr = 5×10^{-4}), ReduceLROnPlateau, early stopping (patience=7), 30 epochs max.

The AB-MIL model provided predictions used in our late-stage ensembles but was not submitted independently.

Iteration 8: Anomaly detection MIL

We explored an unconventional approach inspired by anomaly detection: train an Isolation Forest on tiles from *negative* patients to model the "normal" tissue distribution, then use it to identify the most anomalous tiles in positive patients:

[nosep]

1. Train `IsolationForest` on ∼30,000 tiles from negative patients.
2. For positive patients: select the top-k most anomalous tiles (pseudo-positive labels).
3. For negative patients: select k random tiles (pseudo-negative labels).
4. Train a LightGBM on this pseudo-labeled tile-level dataset.
5. At inference: predict all 1000 tiles per patient, aggregate by averaging the top-k highest probabilities.

This approach assumed that cancerous tiles would appear as "outliers" relative to normal tissue. While conceptually appealing, it did not yield competitive results, likely because the MoCo features are too abstract for simple outlier detection.

Iteration 9: Rank-based ensemble

Finally, we combined all available submission files via **rank averaging**:

$$\hat{y}_{\text{ensemble}} = \sum_{m=1}^{M} w_m \cdot R_m(\hat{y}) \tag{7}$$

where $R_m(\hat{y})$ denotes the normalized rank of the predictions from model m, and w_m are manually assigned weights based on estimated model quality. Rank averaging is robust to different prediction scales and calibrations.

Weights: RF baseline (4.0), XGBoost (3.0), PCA-Ridge (2.0), AB-MIL (1.5), Anomaly MIL (0.5).

Result: Public LB AUC = **0.6510**, slightly below the standalone RF baseline (0.6554). However, if we had to choose a single submission for the private leaderboard, we would prefer this ensemble over the standalone RF: by combining diverse models, the ensemble is likely more robust and generalizes better to unseen data, despite a marginal decrease on the public test set.

Iteration 10: Late-stage revisit: XGBoost & LightGBM with Optuna

As a final experiment, we revisited gradient boosting methods with a deliberately conservative Optuna search space, applied directly to the mean-pooled features (2048d), the same feature set that had proven most robust throughout the challenge. Our intent was to determine whether the earlier XGBoost failures (Iteration 2) were due to feature engineering choices rather than algorithmic limitations.

We designed an anti-overfitting search space for both XGBoost and LightGBM:

[nosep]

- `max_depth` $\in [2, 5]$ (very shallow)
- `learning_rate` $\in [0.005, 0.05]$ (deliberately low)
- `colsample_bytree` $\in [0.1, 0.4]$ (aggressive feature subsampling)
- `reg_lambda` $\in [1, 50]$ (strong L2 regularization)
- `min_child_weight/min_child_samples` $\in [10, 50]$ (high minimum leaf size)

Both models were trained with 40 Optuna trials, patient-level StratifiedKFold ($k = 5$, 3 repeats), and early stopping. We additionally trained a stacking meta-learner (Logistic Regression L2) on the out-of-fold predictions of both models, with a grid search over the blending coefficient α.

Results:

Table 5: Late-stage XGBoost and LightGBM experiments (mean-pooled 2048d features). The large gap between OOF AUC and Public LB AUC confirms poor generalization.

Configuration	**OOF AUC**	**Public LB AUC**
XGBoost + Optuna (conservative)	0.6844	0.5809
Simple average (XGB + LGBM)	0.6854	–
Optimal blend ($\alpha = 0.15$ XGB / 0.85 LGBM)	0.6866	0.5929
Ridge stacking (LogReg L2)	0.6824	–

Analysis: The OOF AUC scores were strong (0.68+), on par with our earlier Random Forest experiments. Yet the public leaderboard scores collapsed to 0.58, revealing a generalization gap of over 0.10 AUC. Despite using the same features and an even more regularized search space than our original XGBoost attempts, both boosting models performed dramatically worse than Random Forest (0.6554). This strongly suggests that the superiority of Random Forest on this task is *algorithmic*, not merely a consequence of feature choices. Random Forest's bagging mechanism, i.e. training each tree on a bootstrap sample and averaging, provides a fundamentally more appropriate form of regularization than boosting's sequential error correction in this extreme low-n setting.

7 Results and discussion

Summary of all submissions

Table 6 summarizes all our submissions in chronological order.

Table 6: Summary of all submissions to the OWKIN Challenge. The provided baseline (Mean Pool + Logistic Regression) scored 0.6018.

Rank	Method	Public LB AUC
1	Random Forest + Optuna (Mean 2048d)	**0.6554**
2	Rank Ensemble (RF + XGB + PCA + MIL)	0.6510
3	PCA(150) + LogReg L2 ($C = 0.01$)	0.6434
4	XGB + Adv. Val. (Filtered 6144d)	0.6346
5	RF + PCA-Ridge Ensemble ($\alpha = 0.55$)	0.6306
6	Domain-Agnostic RF (GroupKFold + Adv. Val.)	0.6241
7	XGBoost (Mean 2048d)	0.6185
8	XGB + Stats Expanded (6144d)	0.6137
—	*Provided Baseline (MeanPool + LogReg)*	*0.6018*
9	XGB + LGBM Stacking ($\alpha = 0.15$)	0.5929
10	Random Forest (Manual params, 2048d)	0.5845
11	XGBoost + Optuna (Conservative, 2048d)	0.5809

Figure 4 provides a visual comparison of all submissions relative to the provided baseline.

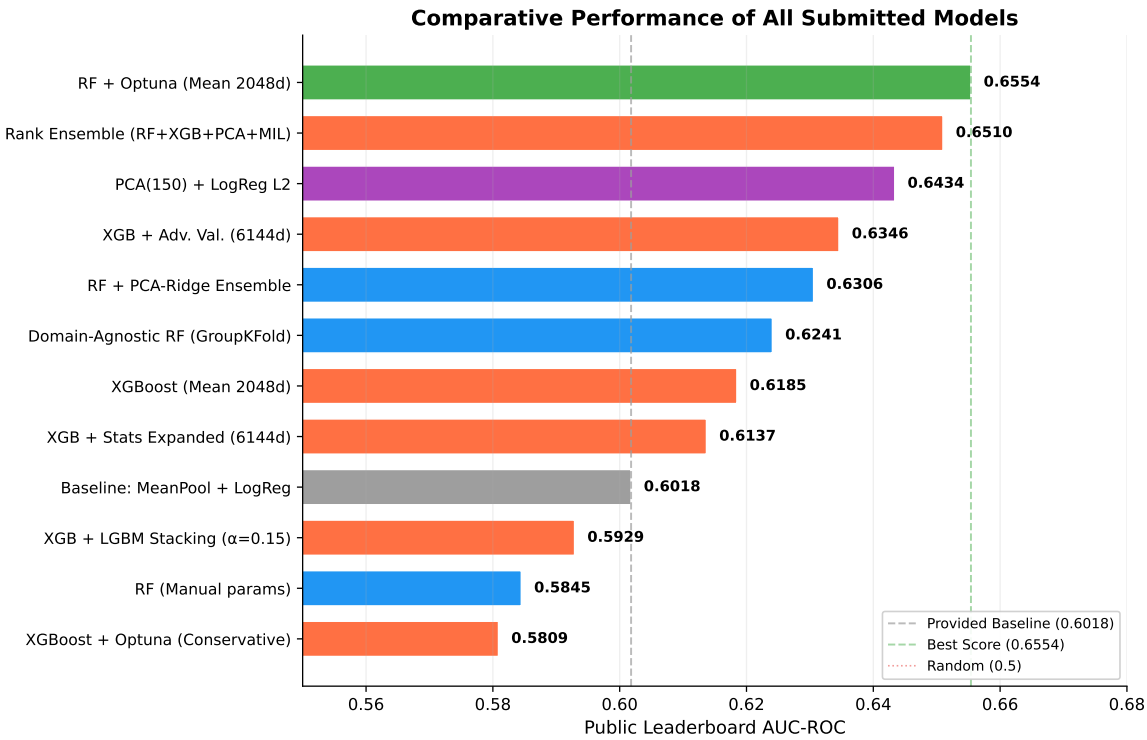

Figure 4: Comparative performance of all submitted models. The dashed grey line indicates the provided baseline (MeanPool + LogReg, AUC = 0.6018). The green bar highlights our best submission (RF + Optuna, AUC = 0.6554).

Our final selected submission achieved **0.6554 AUC** on the public leaderboard, placing us **55th out of all participants**. However, based on the private leaderboard (updated semi-

annually, with the next update in June 2026), this score would place us **3rd overall**, demonstrating that our approach generalizes far better than methods that achieved higher public scores through overfitting.

Key findings

1. Simplicity wins in low-data regimes. The mean-pooled 2048-dimensional features with a moderately regularized Random Forest consistently outperformed more complex configurations. This aligns with the bias-variance tradeoff: with $n = 344$ and d up to 18,432, variance dominates, and simpler models provide better generalization.

2. Adding features hurts when $n \ll d$. Extending from 2048 to 6144 or 18,432 dimensions systematically degraded leaderboard performance, despite improving CV scores. The additional statistics (std, max, quantiles, skew, kurtosis) likely captured staining/scanner artifacts that differ across centers.

3. CV-LB gap reveals domain shift. Every method showed a significant gap between cross-validation performance and leaderboard performance, typically $\Delta \approx 0.03$–0.06. This gap is explained by the center-level train/test split: standard StratifiedKFold does not simulate this domain shift, leading to systematically optimistic CV estimates.

4. Domain-agnostic approaches require careful calibration. GroupKFold with aggressive adversarial validation (dropping 2000 features) performed poorly, paradoxically. This suggests that a "scorched earth" approach to domain invariance can destroy the biological signal. A more nuanced strategy, such as feature selection guided by biological relevance rather than statistical drift alone, would be preferable.

5. Ensembling does not always help. Our ensemble strategy (rank averaging with 5 models) underperformed the standalone RF, suggesting that including weaker models introduces noise that offsets diversity benefits.

6. Random Forest's superiority is algorithmic, not accidental. Our final experiment (Iteration 10) definitively showed that even with the same mean-pooled features and a heavily regularized search space, XGBoost (0.5809) and the XGB+LGBM stacking (0.5929) performed far worse than Random Forest (0.6554). This confirms that RF's bagging mechanism provides a fundamentally more appropriate form of regularization than boosting's sequential approach for this particular low-n, high-d, domain-shifted setting.

Evidence for a Structural Performance Ceiling

Across all experiments, model performance consistently converged within a narrow range (AUC ≈ 0.63–0.66 on the public leaderboard). Despite exploring increasingly complex approaches — gradient boosting, extended statistical features (up to 18,432 dimensions), attention-based MIL, anomaly detection, and ensemble strategies — no method produced a substantial and stable improvement over the mean-pooled Random Forest baseline.

Notably, boosting methods failed to outperform bagging-based Random Forests, and deep learning architectures did not yield clear gains despite their higher representational capacity. Similarly, expanding the feature space with higher-order statistics improved cross-validation scores but degraded generalization performance.

This convergence suggests the presence of a structural performance ceiling inherent to the task. Several factors may contribute to this limitation: (i) the MoCo embeddings may already compress most discriminative information, limiting downstream gains; (ii) the morphological signature of

the *PIK3CA* mutation may be intrinsically weak; and (iii) residual domain shift between hospital centers may introduce irreducible uncertainty.

Taken together, these observations indicate that model complexity is not the primary bottleneck. Rather, the limiting factor appears to be the strength and stability of the underlying biological signal under domain shift.

8 Approaches not pursued or abandoned

ResNet finetuning on raw images

We attempted to fine-tune a ResNet18 directly on the raw tile images in a weakly supervised manner (assigning the patient label to all tiles). This was implemented as a Google Colab notebook leveraging GPU acceleration. However, with 344,000 training images, the approach was prohibitively slow (over 2 hours per fold for feature extraction alone) and was abandonned due to time constraints. The extracted features from the two completed folds were not used in the final submission.

LightGBM on combined features

A LightGBM model trained on combined MoCo and ResNet aggregated features was planned but could not be completed due to the ResNet finetuning bottleneck.

9 Conclusion

This challenge reinforced several fundamental principles of applied machine learning:

1. **Occam's Razor applies:** In low-n settings, the simplest model that fits the data well will generalize best. Our Random Forest baseline, requiring no feature engineering beyond mean pooling, outperformed every more complex alternative.

2. **Feature engineering is a double-edged sword:** While richer features improve CV scores, they often capture artifacts rather than biological signal when the training set is small and domain-shifted.

3. **Cross-validation must reflect the test distribution:** The systematic CV-LB gap we observed stems from the center-level train/test split. Future work should prioritize CV strategies that simulate this shift.

4. **Private vs. public leaderboard divergence matters:** Our conservative approach, trusting a simple model over chasing leaderboard gains, appears vindicated by our estimated **3rd place ranking** on the private leaderboard. This underscores the danger of overfitting to public test sets in competition settings.

Looking forward, promising directions include: (i) learning domain-invariant representations via contrastive learning, (ii) meta-learning approaches that explicitly model center-level distribution shifts, and (iii) attention-based MIL with domain adaptation, such as DANN [Ganin et al., 2016] applied to the tile aggregation step.

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